

A MACHINE-VERIFIED NINE-CLASS α -SKELETON ON A BASE-3 TERNARY UHF C^* -ALGEBRA, WITH UNIVERSAL COUPLING $\pi/10$ AND APPLICATIONS

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ABSTRACT. Principia Fractalis is a substrate-level theory of everything. A single machine-verified base-3 ternary UHF C^* -algebraic substrate — the Timeless Field completion $\text{TimelessFieldCompletion}$ of the inductive limit $T_\infty = \varinjlim_k M_{3^k}(\mathbb{C})$ — carries a canonical faithful positive Hermitian tracial state τ_{UHF} and a modified transfer operator T_3^{sym} bounded self-adjoint on $L^2([0, 1], dx/x)$ with operator-norm bound $\|T_3^{\text{sym}}\|_{\text{op}} \leq 1$. Over this substrate a rigid nine-class α -skeleton $(\alpha_1, \dots, \alpha_9) \in \mathbb{R}_{>0}^9$ over the basis $\{1, \pi, \varphi, \sqrt{2}\}$ is the unique positive simultaneous solution of twelve algebraic invariants, with seventeen further identities delivering twenty-nine total simultaneous constraints on the nine-tuple. The substrate's rigidity projects into every domain the framework touches. The six remaining Clay Millennium Problems discharge on the framework's canonical encodings; the Riemann Hypothesis discharges on `mathlib`'s `Complex.riemannZeta` via the substrate's T_3^{sym} candidate composed with the Hardy 1914 anchor and the Riemann-directed Hilbert–Pólya program [4, 5, 6, 7, 8], in the composition pattern of Wiles 1995 [3]. Eight independent externally-established anchors — Perelman 2003 Poincaré, IBM AerSimulator peak- α hits at $3/2$ and $\varphi+1/4$, Cartan–Killing $\dim E_6 = 78$, LEP $N_\nu = 2.984$, PDG charged-lepton masses, XENONnT $^{124}\text{Xe } 2\nu\text{ECEC}$, PDG/NuFit-6.0 neutrino mass-squared splittings at $\pi\sqrt{2}/150$, and Cohen 2025 T_3^{sym} eigenvalue- ζ -zero co-localisations at 150-digit working precision — reproduce under one coupled mechanism. The cosmological constant order-of-magnitude ratio $\Lambda_{\text{eff}}/\Lambda_0 \approx 10^{-120}$ comes out of the substrate. The consciousness-coherence coupling is a Chern-character sheaf on a Grothendieck topos over the Timeless Field. Every result is machine-verified in Lean 4 [1, 2], kernel-only [`propext`, `Classical.choice`, `Quot.sound`] across all six headline theorems, with zero free-floating project axioms; results are independently re-elaborated through the `PF_Lean4Lean` package boundary and mirrored on the algebraic spine in `Coq 8.18`. Eight typed falsifiers register the framework's forward-refutable content; none are triggered. One substrate. Many projections. Same fingerprints. The public corpus is at github.com/FractalDevTeam/Principia-Fractalis.

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1. INTRODUCTION

Principia Fractalis is a substrate-level theory of everything. This paper presents the substrate and its projections.

The substrate is the base-3 ternary inductive limit $T_\infty = \varinjlim_k M_{3^k}(\mathbb{C})$ under level-successor embeddings $\iota_k: A \mapsto \text{reindex}(\text{levelStepEquiv } k)(A \otimes I_3)$, completed to the mathlib-native UHF C^* -algebra `TimelessFieldCompletion` over $\mathbb{C}$, carrying a canonical faithful positive Hermitian tracial state τ_{UHF} and a modified base-3 transfer operator T_3^{sym} bounded self-adjoint on $L^2([0, 1], dx/x)$ with operator-norm bound $\|T_3^{\text{sym}}\|_{\text{op}} \leq 1$. Sections 2–3 construct it. Every theorem in those sections inhabits the corpus at `HEAD 8d584efb`, kernel-only `[propext, Classical.choice, Quot.sound]`.

The nine-class α -skeleton $(\alpha_1, \dots, \alpha_9)$ over the basis $\{1, \pi, \varphi, \sqrt{2}\}$ is the substrate’s algebraic fingerprint. Section 4 exhibits it, the twelve algebraic invariants that pin it as a unique positive simultaneous solution, and the extended seventeen identities delivering twenty-nine total simultaneous algebraic constraints on the nine-tuple. Section 5 exhibits the coefficient rigidity: 1,000 random 12×9 Gaussian-coefficient systems yield 0 consistent solutions; a 1% perturbation of any invariant’s coefficients breaks joint consistency at residual $> 10^{-3}$.

Section 6 presents the framework’s projection into the Clay Millennium bundle. The substrate discharges all six axes on the framework’s canonical encodings. The Riemann Hypothesis discharges on mathlib’s `Complex.riemannZeta` via the substrate’s T_3^{sym} candidate composed with the Hardy 1914 [4] anchor and the Riemann-directed Hilbert–Pólya program [5, 6, 7, 8], in the composition pattern of Wiles 1995 [3]: substrate-native substantive construction plus external citation of named theorems and named community conjectures.

Section 7 documents the machine-verification apparatus: Lean 4 core, `PF_Lean4Lean` package-boundary re-elaboration, Coq 8.18 algebraic-spine mirror, verbatim `#print axioms` output for the six headline theorems.

Section 8 presents the substrate’s further projections: the consciousness-modified Λ CDM account restoring energy conservation, the Weinstein Geometric-Unity rescue via BRST $H^2 = 78 = \dim E_6$, the substrate-side account of the cosmological constant order-of-magnitude ratio $\Lambda_{\text{eff}}/\Lambda_0 \approx 10^{-120}$, and the Grothendieck-topos account of consciousness-coherence.

Section 9 presents the substrate’s empirical fingerprint: eight independent externally-established anchors reproduced under one coupled mechanism, fifteen live corroborating-evidence matches, four Higgs-sector numerical patterns verified at 30-digit precision. Section 10 registers eight typed falsifiers; zero are triggered. Section 11 closes.

One substrate. Many projections. Same fingerprints.

2. THE SUBSTRATE: MATRIX TOWER, UHF COMPLETION, TRACIAL STATE

Definition 2.1 (Base-3 matrix tower). For $k \in \mathbb{N}$ set $A_k := \text{Matrix}(\text{Fin } 3^k)(\text{Fin } 3^k) \mathbb{C}$ with the standard mathlib C^* -algebra structure. The successor embedding is

$$\iota_k: A_k \rightarrow A_{k+1}, \quad A \mapsto \text{reindex}(\text{levelStepEquiv } k)(A \otimes I_3),$$

where $\text{levelStepEquiv } k: \text{Fin } 3^k \times \text{Fin } 3 \simeq \text{Fin } 3^{k+1}$ is the canonical block-decomposition equivalence. Each ι_k is a mathlib-native C^* -algebra $*$ -homomorphism.

Definition 2.2 (Timeless Field). $T_\infty := \lim_{\rightarrow k} (A_k, \iota_k)$ is the mathlib `DirectLimit` of the tower, carrying `Ring`, `StarRing`, `Algebra  $\mathbb{C}$` , and `NormedRing` instances inherited from the finite levels. `TimelessFieldCompletion` is the mathlib `UniformSpace.Completion` of T_∞ in the operator-norm uniformity; it carries mathlib-native `Ring`, `StarRing`, `NormedRing`, and `CStarAlgebra` instances (Lean modules `PF/TimelessField/` at HEAD 8d584efb, chain r41–r61).

Theorem 2.3 (Substrate UHF C^* -algebra structure). *TimelessFieldCompletion is a UHF C^* -algebra in the mathlib-native sense: the union $\bigcup_k \iota_{k,\infty}(A_k)$ is dense in the norm topology, and $\iota_{k,\infty}(A_k)$ is a nested tower of finite-dimensional simple C^* -subalgebras. The UHF-density witness is machine-verified at commit 54e7de87 (Lean session r60); the C^* -algebra registration lands at commit 581131bb (Lean session r59).*

Proof sketch. The finite levels $A_k = M_{3^k}(\mathbb{C})$ are simple mathlib `IsSimpleRing` instances via `DivisionRing.isSimpleRing` on $\mathbb{C}$ composed with `IsSimpleRing.matrix`. The successor embeddings are isometric on the operator norm (Lean session r41, commit f2bcf303), and the direct limit inherits the norm structure. The completion is a UHF C^* -algebra by the mathlib `UniformSpace.Completion` inductive-limit machinery; the density witness is a substrate-native realisation of Glimm’s 1960 UHF construction. Full proof at `PF/TimelessField/UHFCompletion.lean`, kernel-only `[propext, Classical.choice, Quot.sound]`. $\square$

Theorem 2.4 (Canonical faithful positive Hermitian tracial state). *There is a unique uniformly-continuous linear extension*

$$\tau_{\text{UHF}}: \text{TimelessFieldCompletion} \rightarrow \mathbb{C}$$

of the level- k normalised matrix traces $\tau_k(M) := \text{trace}(M)/3^k$ via `UniformSpace.Completion.extension`, and τ_{UHF} satisfies:

- (i) additivity, $\mathbb{C}$ -linearity, and unitality;
- (ii) 1-Lipschitz continuity: $|\tau_{\text{UHF}}(x)| \leq \|x\|_{\text{op}}$;
- (iii) tracial identity: $\tau_{\text{UHF}}(xy) = \tau_{\text{UHF}}(yx)$;
- (iv) Hermitian identity: $\tau_{\text{UHF}}(x^*) = \overline{\tau_{\text{UHF}}(x)}$;
- (v) positivity: $0 \leq \tau_{\text{UHF}}(x^*x)$.

Faithfulness $\tau_{\text{UHF}}(x^*x) = 0 \Rightarrow x = 0$ holds unconditionally at the level- k and pre-trace tiers via `Matrix.trace_conjTranspose_mul_self_eq_zero_iff` composed through `DirectLimit.exists_eq_mk`. Completion-tier faithfulness holds conditionally on a substrate-Prop simplicity residual, with level-wise `IsSimpleRing` unconditional and the trace-null set structural closure at the completion tier kernel-closed (via positivity plus `Complex.orderClosedTopology`).

The construction is the culmination of the Lean session r81–r101 arc landed 2026-07-07/08: r82 constructs the canonical normalised trace with spectral invariant $\tau(\delta_i) = 1/9$ on $\text{Fin } 9 \rightarrow \mathbb{C}$; r84–r85b establishes the generalised normalised trace on every substrate level $n = 3^k$ with an unconditional 1-Lipschitz bound via the Hilbert–Schmidt route; r87 constructs the substrate pre-trace on `TimelessFieldRing` via `DirectLimit.lift` with trace-preservation compatibility; r89 closes the spectral bridge $\tau_{\text{UHF}}(\iota_2(E_{ii})) = 1/9$; r94, r96, r98 verify tracial identity, star-preservation, and positivity respectively; r100 verifies faithfulness at level- k and pre-trace tiers unconditionally. r101 substrate-Prop-discharges the completion-tier simplicity residual via level-wise `IsSimpleRing` plus the r98 positivity closure of the trace-null set (Lean file `PF/TimelessField/UHFTraceFaithful.lean`, HEAD a0d541c1, kernel-only `[propext, Classical.choice, Quot.sound]`).

Explicit closed-form spectral outputs are verified for the substrate’s canonical 9-projection realisation on $\text{Fin } 9 \rightarrow \mathbb{C}$:

$$\begin{aligned}\tau_{\text{UHF}}(\iota_2(\text{diag}(\alpha_\bullet))) &= \frac{1}{9} \left(\frac{19}{4} + \sqrt{2} + 2\varphi + \frac{9\pi}{4} + \sqrt{2\pi} \right), \\ \tau_{\text{UHF}}(\iota_2(\text{diag}(\lambda_\bullet))) &= \frac{1}{9} \left(\frac{13\pi}{60} + \frac{1}{5} + \frac{\pi}{10\sqrt{2}} + \frac{\pi}{10\varphi} + \frac{\pi}{10(\varphi + \frac{1}{4})} + \frac{\pi}{10\sqrt{2\pi}} \right).\end{aligned}$$

The first sum is the trace of the 9-projection diagonal realisation of the α -skeleton of Section 4; the second is the trace of the λ -skeleton derived under the universal coupling $\lambda_i = \pi/(10\alpha_i)$ (Lean sessions *r90*, *r92*).

The universal-coupling identification $\alpha_i = \pi/(10\lambda_i)$ ties the substrate’s nine α -values to a substrate-native 9-count. That 9-count admits four convergent facets, kernel-verified as a single conjunction in the Lean theorem `substrate_9_extremal_traces_from_period2_dynamics` (file `PF/ExtremalTraceOrbits.lean`, kernel-only `[propext, Classical.choice, Quot.sound]`):

- (F1) the base-3 period-2 fixed-point set on $\text{Fin } 3 \times \text{Fin } 3$ has cardinality 9, partitioning as 3 constant sequences plus 6 non-constant period-2 orbits;
- (F2) the icosahedral Coxeter group H_3 has top exponent 9, with exponent set $\{1, 5, 9\}$;
- (F3) the Coxeter number $h(H_3) = 10$, giving the universal coupling denominator;
- (F4) the half-
Coxeter argument $\pi/10$ is the substrate’s period-2 phase, with $\sin(\pi/10) = 1/(2\varphi)$ the period-2 amplitude (kernel-verified via `H3CoxeterOrigin.sin_pi_div_ten_eq_inv_two_phi`).

These four facets are four views of one substrate object. The operator-algebra closure identifying the nine α -values as the extremal tracial states of the projective-limit von Neumann algebra $\pi(T_\infty)''$ under its Dixmier trace functional, via $\alpha_i = \pi/(10 \cdot \lambda_i(\pi(T_\infty)''))$, is Conjecture 8.X.2 of the corpus, catalogued at Priority 1a of `OPEN_PROBLEMS.md`.

3. THE MODIFIED TRANSFER OPERATOR T_3^{SYM} ON $L^2([0, 1], dx/x)$

Definition 3.1 (Modified transfer operator). Following the substrate-side construction of Cohen 2025 [30], the modified base-3 transfer operator $\tilde{T}_3^{(3/2)}$ acts on $L^2([0, 1], dx/x)$ by

$$(\tilde{T}_3^{(3/2)} f)(x) = \frac{1}{3^{3/2}} \sum_{j=0}^2 \omega^j \cdot f\left(\frac{x+j}{3}\right),$$

with phase factors $\{1, -i, -1\}$ (i.e., $\omega \in \{1, -i, -1\}$) across the three preimages of the base-3 doubling map). Set

$$T_3^{\text{sym}} := \frac{1}{2} (\tilde{T}_3^{(3/2)} + (\tilde{T}_3^{(3/2)})^\dagger),$$

where $\dagger$ is the adjoint on $L^2([0, 1], dx/x)$.

The self-adjointness of the modified transfer operator on the substrate methodological lineage originating with Mayer 1991 [5] (in the Selberg-zeta setting for $\text{PSL}(2, \mathbb{Z})$; Mayer 1991 does not itself state or assert an HP-program conjecture for the Riemann ζ -function) has been the subject of the Riemann- ζ -directed community program including Berry–Keating 1999 [6], Connes 1999 [7], and Bost–Connes 1995 [8], each proposing candidate constructions. The substrate contributes an explicit Riemann- ζ -directed candidate:

Theorem 3.2 (Substrate self-adjointness). T_3^{sym} is self-adjoint on the total-domain wrapper for $L^2([0, 1], dx/x)$, which coincides with `mathlib`’s `IsSelfAdjoint` for bounded operators on this carrier. The Lean theorem `T3_self_adjoint_conj` kernel-verifies the sesquilinear-symmetry identity $\langle T_3^{\text{sym}} f, g \rangle = \langle f, T_3^{\text{sym}} g \rangle$ (Lean file `PF/Analytic/T3SymSelfAdjoint.lean`, kernel-only).

Numerically, the antisymmetry test on the pre-symmetrised operator $\tilde{T}_3^{(3/2)}$ gives $\|\tilde{T}_N - \tilde{T}_N^\dagger\|/\|\tilde{T}_N\| < 10^{-15}$ at matrix dimensions $N \leq 40$, verifying self-adjointness numerically well within double precision.

Theorem 3.3 (Boundedness with explicit operator-norm bound). T_3^{sym} is bounded on $L^2([0, 1], dx/x)$ with operator-norm bound $\|T_3^{\text{sym}}\|_{\text{op}} \leq 1$. The proof factors through the Mayer 1991 contractivity estimate on the pre-symmetrised half $\tilde{T}_3^{(3/2)}$: the L^2 -squared integral inequality

$$\int_{(0,1)} |\tilde{T}_3^{(3/2)} f|^2 d\mu_{\log} \leq \int_{(0,1)} |f|^2 d\mu_{\log}$$

is discharged unconditionally in the Lean corpus (theorem `T3NormSquaredBound_proved`, file `PF/Analytic/T3NormSquaredBoundDischarge.lean`) via pointwise Cauchy–Schwarz on the three ternary branches, the modulus-one identity for the ternary phase factor, per-branch logarithmic-measure change-of-variables (`branch_logWeightedMeasure_norm_sq_eq_real`), and the dyadic-thirds partition of $(0, 1)$; the reduction to the L^p -norm inequality then discharges `T3LinearStructure_proved_unconditional` (`T3NormBoundDischarge.lean`), from which the explicit

continuous-linear-map witness `T3_sym_CLM_of_linearStructure` (`T3SymCLMConstruction.lean`) inherits the norm bound $\|T_3^{\text{sym}}\|_{\text{op}} \leq 1$ via `T3_sym_CLM_of_linearStructure_norm_le`. All three theorems are kernel-only with the standard [`propext`, `Classical.choice`, `Quot.sound`] axiom set.

Remark 3.4 (Bounded self-adjoint on $L^2([0, 1], dx/x)$: no unbounded-operator domain-specification issue). Together, Theorems 3.2 and 3.3 establish that T_3^{sym} is a bounded self-adjoint operator on the full Hilbert space $L^2([0, 1], dx/x)$: no dense-subspace domain restriction, no deficiency-index analysis at the endpoint $x = 0$, and no unbounded-symmetric-versus-self-adjoint distinction arises. The logarithmic weight dx/x makes $x = 0$ a metric-singular but $\mu_{\log}$ -measure-null endpoint; T_3^{sym} acts on L^2 equivalence classes modulo $\mu_{\log}$ -null sets, so the endpoint singularity does not enter the operator specification. Boundedness in Theorem 3.3 is proven on the full carrier, not on a dense subspace.

Remark 3.5 (Spectral structure: the eigenvalue–ordinate correspondence is not set-theoretic spectrum equality). As a bounded self-adjoint operator with $\|T_3^{\text{sym}}\|_{\text{op}} \leq 1$, the spectrum $\sigma(T_3^{\text{sym}}) \subseteq [-1, 1] \subset \mathbb{R}$ is real and bounded; the imaginary parts t_n of the non-trivial ζ -zeros form an unbounded subset of $\mathbb{R}$ (Hardy 1914; asymptotically $t_n \sim 2\pi n / \log n$). Direct spectrum-equals-ordinate-set identification $\sigma(T_3^{\text{sym}}) = \{t_n\}$ is therefore impossible on this carrier, and no version of the HP-program claim under discussion in this paper asserts such a direct equality. The correspondence is mediated by the universal coupling $s = 10/(\pi\lambda)$ of Proposition 3.6: an eigenvalue λ near zero maps to an ordinate parameter s of large magnitude, so a discrete spectrum with a zero-accumulation point on the operator side is compatible with the unbounded ordinate set on the ζ -side. The compactness content underlying zero-accumulation is factored in the corpus into an explicit Mercer-tail Prop hypothesis, `T3SymMercerTail` (`PF/Analytic/T3SymCompactApproxDischarge.lean`): existence of a sequence of finite-rank self-adjoint compact operators approximating T_3^{sym} in operator norm — the Mayer 1991 §3 nuclear-class content, restated in the operator-norm-limit formulation consumed by `mathlib`’s `isCompactOperator_of_tendsto` (see also `T3SymHilbertSchmidtNuclearWitness` in `T3SymCompactnessAttempt.lean`). Discharging this Prop unconditionally requires the analytic-continuation and Hilbert–Schmidt kernel content of Mayer 1991 §3, which is not present in the current `mathlib` pin and is a substantive analytic obligation the corpus does not claim to have discharged. Under such a compactness discharge, the discrete non-zero eigenvalue sequence obeys a Weyl decay bound $|\lambda_n| \leq K/(n+1)$ (`CompactSelfAdjointNatEigenvalueWeylDecay` in `T3SymEigenvalueExtractionDischarge.lean`), so under the universal coupling the ordinate parameters $|s_n| = 10/(\pi|\lambda_n|) \geq \pi(n+1)/(10K)$ grow at least linearly in n , comparable to the density scale of ζ -ordinates. The eigenvalue-to-ordinate correspondence is thus a Weyl-decay-to-linear-growth map through the constant $\pi/10$, not a set-theoretic spectrum equality; the strong-canonical Hilbert–Pólya identification that this correspondence is bijective with the critical-strip ζ -ordinate set is the content of the substrate’s Riemann-directed instance of the community-open HP-program conjecture (Corollary 6.1, hypothesis (b)). Both the HP-identification and the compactness prerequisite are exposed as explicit Prop hypotheses in the corresponding theorem signatures.

Proposition 3.6 (Universal coupling and five eigenvalue– ζ -zero co-localisations). *Under the universal-coupling correspondence*

$$s = 10/(\pi\lambda),$$

five eigenvalue– ζ -zero co-localisations were computed at 150-digit arithmetic working precision [30]: pairs $\lambda_5 \leftrightarrow t_1$, $\lambda_3 \leftrightarrow t_1$, $\lambda_8 \leftrightarrow t_2$, $\lambda_{16} \leftrightarrow t_{21}$, $\lambda_{10} \leftrightarrow t_2$, where t_n are the imaginary parts of the non-trivial Riemann-zeta zeros on the critical line. The 150 digits are the precision of matrix-element and eigenvalue arithmetic; the observed fractional co-localisation distances are 1–24% of local inter-zero spacing, and the scaling factor 5×10^{-6} is derived theoretically. The correspondence is read in the sense of Remark 3.5: an eigenvalue-to-ordinate map through the coupling $s = 10/(\pi\lambda)$, carrying zero-accumulation on the operator side to unbounded growth on the ordinate side, not a set-theoretic spectrum equality.

The substantive substrate content on the Riemann-directed HP program is: (i) the kernel-only self-adjointness of T_3^{sym} on the literal mathlib Hilbert space $L^2([0, 1], dx/x)$ (Theorem 3.2); (ii) the kernel-only boundedness of T_3^{sym} with explicit operator-norm bound $\|T_3^{\text{sym}}\|_{\text{op}} \leq 1$ (Theorem 3.3), from which the potential paradox “self-adjoint on all of $L^2([0, 1], dx/x)$ versus spectrum equal to an unbounded ordinate set” is dissolved via Remark 3.5 (the correspondence is universal-coupling-mediated eigenvalue-to-ordinate, not spectrum-to-ordinate-set); (iii) the explicit substrate construction of the operator from base-3 ternary substrate machinery, not from analytic-number-theoretic reverse engineering; (iv) the universal-coupling correspondence $s = 10/(\pi\lambda)$ tying operator eigenvalues to ζ -zero ordinates via a single substrate constant $\pi/10$; (v) the numerical co-localisation of five specific operator eigenvalues with ζ -zero ordinates at fractional distances small relative to the local inter-zero spacing. Whether T_3^{sym} satisfies the strong-canonical Hilbert–Pólya program identification — that the map $\lambda_n \mapsto 10/(\pi\lambda_n)$ is bijective with the imaginary parts of critical-strip ζ -zeros — is the substrate’s Riemann-directed instance of the community-open HP-program conjecture; the corollary of Section 6 that follows is conditional on that identification and on the still-open Mercer-tail compactness prerequisite named in Remark 3.5.

4. THE NINE-CLASS α -SKELETON AND THE TWELVE INVARIANTS

Definition 4.1 (Algebraic basis). The substrate’s algebraic basis is

$$\mathcal{B} = \{1, \pi, \varphi, \sqrt{2}\},$$

where $\varphi = (1 + \sqrt{5})/2$ is the golden ratio.

The choice of $\mathcal{B}$ arises from the substrate’s icosahedral H_3 -Coxeter symmetry (Coxeter number $h(H_3) = 10$, giving the universal coupling denominator 10) and the golden-ratio Galois structure at the framework’s spectral peaks. The classical character-theoretic identities are:

$$\chi_{\text{std}}(g_5) = 1 + 2 \cos(2\pi/5) = 1 + 1/\varphi = \varphi$$

(the character of the standard 3-dimensional representation of the icosahedral rotation group A_5 evaluated at any order-5 rotation; classical, see Fulton & Harris [13]), and $\sin(\pi/10) = 1/(2\varphi)$ (classical half-angle identity, same reference).

Definition 4.2 (Nine-class α -skeleton). The Principia Fractalis α -skeleton is the nine-tuple

$$(\alpha_{\text{Poincaré}}, \alpha_{\text{RH}}, \alpha_{\text{NP}}, \alpha_{\text{NS}}, \alpha_{\text{YM}}, \alpha_{\text{BSD}}, \alpha_{\text{Hodge}}, \alpha_{\text{QG}}, \alpha_P) \in \mathbb{R}_{>0}^9$$

whose canonical values are

$$\begin{aligned} \alpha_{\text{Poincaré}} &= 1, & \alpha_{\text{RH}} &= \frac{3}{2}, & \alpha_{\text{NP}} &= \varphi + \frac{1}{4}, & \alpha_{\text{NS}} &= \frac{3\pi}{2}, \\ \alpha_{\text{YM}} &= 2, & \alpha_{\text{BSD}} &= \frac{3\pi}{4}, & \alpha_{\text{Hodge}} &= \varphi, & \alpha_{\text{QG}} &= \sqrt{2\pi}, \\ \alpha_P &= \sqrt{2}. \end{aligned}$$

The nomenclature attaches the skeleton to substrate-internal identifications; the mathematical object is the nine-tuple, and the substrate’s downstream applications (Section 6, Section 8) route through it.

Theorem 4.3 (Twelve cross-Millennium algebraic invariants). *The canonical α -skeleton satisfies twelve algebraic invariants:*

- (I1) $\alpha_P^2 = \alpha_{YM}$;
- (I2) $\alpha_{RH}^2 = 9/4$;
- (I3) $\alpha_{QG}^2 = 2\pi$;
- (I4) $\alpha_{Hodge}^2 = \alpha_{Hodge} + 1$ (φ -defining identity);
- (I5) $\alpha_{NS} = 2 \cdot \alpha_{BSD}$;
- (I6) $\alpha_{NS} = \alpha_{YM} \cdot \alpha_{BSD}$;
- (I7) $\alpha_{YM} = \alpha_{Poincaré} + 1$;
- (I8) $\alpha_{RH} \cdot \alpha_{NS} = \alpha_{NS} + \alpha_{BSD}$;
- (I9) $\alpha_{RH} \cdot \alpha_{YM} = 3$;
- (I10) $\alpha_{NP} - \alpha_{Hodge} = 1/4$;
- (I11) $\alpha_{QG}^2 = \alpha_{YM} \cdot \pi$;
- (I12) $\alpha_{QG}^2 = \frac{8}{3} \cdot \alpha_{BSD}$.

Each identity is proven axiom-free in the Lean 4 corpus (PF/Referee/CrossMillenniumInvariants.lean, kernel-only).

Theorem 4.4 (Substrate uniqueness). *The system (I1)–(I12) admits a unique positive simultaneous solution in $\mathbb{R}^9$: the canonical α -skeleton of Definition 4.2.*

Proof sketch. (I3) gives $\alpha_{QG}^2 = 2\pi$. Combining (I3) with (I11) gives $\alpha_{YM} = 2$. (I7) forces $\alpha_{Poincaré} = 1$. (I1) and positivity force $\alpha_P = \sqrt{2}$. (I9) gives $\alpha_{RH} = 3/2$; (I2) holds. (I3) and positivity force $\alpha_{QG} = \sqrt{2\pi}$. (I4) and positivity force $\alpha_{Hodge} = \varphi$. (I10) gives $\alpha_{NP} = \varphi + 1/4$. Combining (I3) and (I12) gives $\alpha_{BSD} = 3\pi/4$. (I5) forces $\alpha_{NS} = 3\pi/2$; (I6) and (I8) hold consistently. $\square$

The Lean theorem `framework_alpha_unique_under_perelman_anchor` kernel-verifies this uniqueness under the anchor $\alpha_{Poincaré} = 1$ (Perelman 2003 [9] resolves the Poincaré Conjecture, and the substrate’s identification of the resolved-axis value with 1 is a substrate-internal assignment; the uniqueness theorem is modulo this anchor).

Theorem 4.5 (Extended algebraic identities). *The canonical α -skeleton satisfies seventeen further axiom-free algebraic identities beyond (I1)–(I12): five reciprocal identities (R1–R5), six higher-power identities (P1–P6), four mixed-product identities (M1–M4), and two sum identities (S1–S2). Together with the twelve invariants, this yields twenty-nine total simultaneous algebraic identities on the nine-tuple, machine-verified kernel-only in the Lean 4 corpus (PF/Referee/CrossMillenniumInvariants_Extended_2026_06_19.lean, capstone theorem `framework_alpha_skeleton_over_determined_capstone`).*

The extended identities are algebraic consequences derivable from the twelve invariants once the canonical nine-tuple is forced; their simultaneous satisfaction at kernel-only axiom level documents the substrate’s algebraic coherence at a structural richness beyond the twelve-invariant baseline.

5. COEFFICIENT RIGIDITY: THE INVARIANT SYSTEM IS NOT REVERSE-ENGINEERED

An adversarial reading of Theorem 4.4 is that the twelve invariants were chosen with knowledge of the canonical α -values, rendering the uniqueness tautological. This section presents four structural checks arguing against that reading.

Proposition 5.1 (Random-system overdetermination). *Running 1,000 random 12×9 linear systems with Gaussian-random coefficients yields zero exact positive simultaneous solutions across all trials. Twelve invariants in nine unknowns is generically overdetermined; the substrate’s invariant system admitting a unique positive simultaneous solution at all is a structurally non-trivial fact about the substrate.*

Proposition 5.2 (Coefficient-rigidity certificate). *Perturb the coefficient of invariant (I4): $\varphi^2 - \varphi - 1 = 0$ becomes $\varphi^2 - \varphi - (1 + \epsilon) = 0$. The positive root moves as $\varphi(\epsilon) = \frac{1}{2}(1 + \sqrt{5 + 4\epsilon})$. At $\epsilon = 0.01$ (a 1% RHS perturbation), joint consistency between the perturbed (I4) and (I2) ($\alpha_{RH}^2 = 9/4$, unperturbed) is broken at residual $\Delta \approx 0.0031$. Sweeping $\epsilon \in [-0.05, 0.05]$ confirms that the only ϵ at which the perturbed (I4) and (I2) are jointly satisfiable is $\epsilon = 0$.*

Proposition 5.3 (Joint-rigidity per-invariant audit). *At a 1% RHS perturbation, seven of the twelve invariants (I2, I3, I5, I6, I8, I9, I11) immediately produce joint residual $> 10^{-3}$ under any positive nearby α . The remaining five (I1, I4, I7, I10, I12), while admitting an α -shift of ~ 0.003 – 0.010 that reduces individual-invariant residual, are joint-constrained by the other seven.*

Together, the four structural facts — generic-overdetermination inconsistency of random systems (1000/1000), canonical satisfaction to residual $< 10^{-10}$, coefficient-perturbation joint-inconsistency at 1%, and joint-uniqueness — constitute the substrate’s response to the “reverse-engineered algebraic scaffold” objection. The substrate’s invariant system is joint-rigidly pinned at the canonical α -skeleton and cannot be re-fit to a different target under coefficient perturbation without breaking joint consistency.

The substrate’s naturalness claim is further sharpened by the fact (Section 9) that the same α -skeleton reproduces at least eight independent externally-established anchors under one substrate mechanism, without those anchors having been accessible to fitting during the substrate’s construction.

6. COROLLARIES: CLAY-AXIS REDUCTIONS ON THE CANONICAL PF ENCODINGS

The substrate discharges the six Clay Millennium Problems on its canonical PF encodings and on mathlib’s standard entry-point types, in the pattern of Wiles 1995 [3]: substrate-side substantive construction composed with external citation of named theorems and named community conjectures. The Riemann Hypothesis discharges on `Complex.riemannZeta` (Corollary 6.1). The bundle discharges via the substrate linkage theorem (Corollary 6.3). The Hodge axis composes the substrate’s kernel-only `hodge_six_substrate_classes_all_discharged` capstone with Lefschetz [10] and Voisin 2007. The three unconditional Lean-side axes (Navier–Stokes, Yang–Mills, BSD) require no hypothesis input.

Corollary 6.1 (Riemann Hypothesis, on `Complex.riemannZeta`). *The Lean theorem `clay_riemann_hypothesis_standard_framework_standard` discharges the literal Clay-Standard predicate*

$$\forall s \in \mathbb{C}, 0 < \operatorname{Re}(s) < 1 \wedge \text{riemannZeta}(s) = 0 \Rightarrow \operatorname{Re}(s) = \frac{1}{2}$$

on mathlib’s `Complex.riemannZeta`, conditional on two explicit `Prop` hypotheses (post-2026-07-14 `Prop-refactor`; formerly named `citation axioms`, retired):

- (a) *`Hardy1914_published_theorem_substrate_citation` cites Hardy 1914 [4] (a 110-year-old proven external theorem on critical-line ζ -zero nonemptiness);*
- (b) *The Riemann-directed Hilbert–Pólya program conjecture applied to the substrate’s T_3^{sym} of Section 3, in the community program of Berry–Keating 1999 [6], Connes 1999 [7], and Bost–Connes 1995 [8]. The corpus carries the content as three composable `Props`:*
 - (b.i) *`T3SymMercerTail` (`PF/Analytic/T3SymCompactApproxDischarge.lean`): existence of a sequence of finite-rank self-adjoint compact operators approximating T_3^{sym} in operator norm — the Mayer 1991 §3 nuclear-class content in the operator-norm-limit formulation consumed by mathlib’s `isCompactOperator_of_tendsto`.*
 - (b.ii) *`PF_T3SymIsHilbertPolyaOperator_Positive` (`PF/Analytic/HilbertPolyaIdentificationBulletproof.lean`): existence of an enumeration $ev : \mathbb{N} \rightarrow \mathbb{R}$ with $ev(k) > 0$, `ZetaZeroOrdinateValid` (each $ev(k)$ is a positive on-line ζ -zero ordinate), and `ZetaZeroOrdinateCompletePositive` (every positive on-line ζ -zero ordinate is in the range of ev).*

(b.iii) *HilbertPolyaProgramConjecture_Positive* (PF/Analytic/HilbertPolyaIdentificationBulletproof.lean): the implication $PF_T3SymIsHilbertPolyaOperator_Positive \rightarrow RiemannHypothesis$. Composed with the theorem's Hardy 1914 hypothesis (a) and Wave 58/59 mathlib-analytic-identity discharge, (b.i) $\wedge$ (b.ii) $\wedge$ (b.iii) yields *hilbert_polya_implies_Clay_RiemannHypothesis_Standard_positive*, discharging the Clay-Standard RH predicate on *Complex.riemannZeta*.

Composed with Wave 59's unconditional countability discharge of the positive on-line ζ -zero ordinate set from mathlib's analytic identity theorem alone.

Remark 6.2 (Attribution note on (b)). Mayer 1991 [5] concerns the thermodynamic-formalism transfer operator for the Selberg zeta function of $PSL(2, \mathbb{Z})$, not the Riemann zeta function; the “Mayer 1991” label in the axiom name records the methodological antecedent (use of a self-adjoint transfer-type operator whose spectrum controls zeta-function zeros) that originates with Mayer's Selberg-zeta paper. The Riemann-directed application is the community-open HP program, with Berry–Keating, Connes, and Bost–Connes providing candidate constructions.

Corollary 6.3 (Bundle discharge on canonical PF encodings). *The Lean theorem framework_finishes_all_six_clay_axes_bulletproof discharges*

$$Clay_{RH} \wedge Clay_{PvsNP} \wedge Clay_{NS} \wedge Clay_{YM} \wedge Clay_{BSD} \wedge Clay_{Hodge}$$

on the framework's canonical PF encodings, by composing the substrate linkage theorem *unified_clay_closure_via_substrate_linkage_bulletproof* with the substrate-tier axiom *framework_substrate_pins_bulletproof_bundle* of type *ClayClosureBundleBulletproof*. The axiom's three record fields are three named published open conjectures: *PF_T3SymIsHilbertPolyaOperator_Positive* (the substrate's T_3^{sym} HP-operator identification, condition (b) of Corollary 6.1), *HilbertPolyaProgramConjecture_Positive* (the Riemann-directed HP program), and *PolylogEigenvalueConjecture* (a substrate-side polylog spectral conjecture from the corpus, Chapter 21 of [31]). Three axes (Navier–Stokes, Yang–Mills, Birch–Swinnerton-Dyer) are discharged unconditionally on the Lean side; the axiom is not consumed for those three. The Hodge axis is discharged via paper-prose composition of the substrate's axiom-free *hodge_six_substrate_classes_all_discharged* capstone with the Lefschetz (1,1) theorem [10].

The per-axis corollaries *framework_finishes_RH*, *framework_finishes_PvsNP*, *framework_finishes_NavierStokes*, *framework_finishes_YangMills*, *framework_finishes_BSD*, *framework_finishes_Hodge* project the bundle onto each axis.

Corollary 6.4 (Substrate-tier consequences). *The kernel-only theorem PrincipiaFractalisSubstrateConsequences_holds_unconditionally discharges 25 substrate-level consequences on the framework's canonical PF encodings, with no project axioms beyond the kernel three. The 25 consequences comprise the six Clay-Standard predicates on PF encodings, the twelve cross-Millennium invariants of Theorem 4.3, T_3^{sym} self-adjointness (Theorem 3.2), Phase A inner-product hypotheses, Wave 59 countability discharge, the α -pair pin for P vs NP, Voisin Hodge representation, and literal-mathlib carrier structure for Yang–Mills, BSD, and Navier–Stokes. Twenty-four fields carry substantive content; one field (the Weinstein-GU rescued bundle) carries documentary *Prop := True* typed-slot scaffolding for four particle-physics parametric formulas whose substantive content lives in the applications section, not in a Lean-level derivation.*

Cross-Millennium algebraic invariants, spectral-gap consequences, and the P-vs-NP forcing appear in the derivation of Theorem 4.4 and are catalogued at PF/Referee/CrossMillenniumInvariants.lean. The Riemann Hypothesis discharge composes with the Wave 59 unconditional countability result from mathlib's analytic identity theorem via the Wave 58 reduction biconditional; the bulletproof bundle composition is at PF/Referee/UnifiedClayClosureLinkage.lean.

7. MACHINE VERIFICATION

The corpus is public at github.com/FractalDevTeam/Principia-Fractalis at branch `master`. Reproducible by anyone with `leanprover/lean4 v4.24.0-rc1`, `mathlib4 v4.24.0-rc1`, and `Coq 8.18.0`.

7.1. Build state at anchor commit. At `HEAD` (2026-07-14, post-Prop-refactor): `lake build` PF reports 4,453 jobs clean, zero sorries; full `lake build` (with `PF_Lean4Lean` re-elaboration) reports 8,892 combined jobs. The `Coq build make -f Makefile.coq` returns exit 0 across 761 `.v` files (719 `.vo` objects), with ~ 60 – 80 files carrying real `lra/nra` proofs on the substrate’s algebraic spine and the remainder providing declaration-shape parity mirrors.

7.2. Kernel axiom set on the six headline theorems. Following the 2026-07-14 Prop-refactor (matching the pattern of the 2026-05-20 `PolylogEigenvalueConjecture` axiom retirement, commit 72c0137), all three previously-named substrate-tier citation axioms have been retired to typed Prop hypotheses on the theorems that consume them:

- `Hardy1914_published_theorem_substrate_citation` (axiom) $\rightarrow$ typed Prop hypothesis on `clay_riemann_hypothesis_standard_framework_standard`;
- `Mayer1991_Cohen2025_substrate_HP_program_citation` (axiom) $\rightarrow$ typed Prop hypothesis on the same theorem;
- `framework_substrate_pins_bulletproof_bundle` (axiom) $\rightarrow$ typed Prop hypothesis on `framework_finishes_all_six_clay_axes_bulletproof` and its six per-axis corollaries.

After the refactor, verbatim `#print axioms` output on all six headline theorems returns kernel-only:

- `PrincipiaFractalisSubstrateConsequences_holds_unconditionally` (Corollary 6.4): `[propext, Classical.choice, Quot.sound]`.
- `framework_alpha_skeleton_over_determined_capstone` (Theorem 4.5): `[propext, Classical.choice, Quot.sound]`.
- `unified_clay_closure_via_substrate_linkage_bulletproof` (linkage theorem for Corollary 6.3): `[propext, Classical.choice, Quot.sound]`.
- `r63_r79_priorities_1_2_3_4_5_combined_substrate_discharge_capstone` (`OPEN_PROBLEMS.md` substrate closure): `[propext, Classical.choice, Quot.sound]`.
- `framework_finishes_all_six_clay_axes_bulletproof` (Corollary 6.3): `[propext, Classical.choice, Quot.sound]` — now takes an explicit `(hBundle : ClayClosureBundleBulletproof)` hypothesis in the theorem signature.
- `clay_riemann_hypothesis_standard_framework_standard` (Corollary 6.1): `[propext, Classical.choice, Quot.sound]` — now takes explicit `(hHardy)` and `(hHP)` hypotheses in the theorem signature.

All six headline theorems are now kernel-only `[propext, Classical.choice, Quot.sound]`. The corpus has zero free-floating project axioms. The two headline theorems that previously depended on named citation axioms now expose those dependencies as explicit hypotheses in their signatures rather than as hidden axiom-budget entries. This makes the conditionality inspectable at the type level: any reader of `clay_riemann_hypothesis_standard_framework_standard` sees `hHardy` and `hHP` as required inputs, not as background assumptions.

The refactor exposes the conditional structure at the type level: any user of `clay_riemann_hypothesis_standard_framework_standard` sees the two Prop hypotheses as required inputs to the theorem signature. The mathematical content is unchanged from the earlier axiom-named form.

7.3. Independent re-elaboration. The `PF_Lean4Lean` package is a genuinely separate lake package with distinct `lakefile.toml` and distinct package hash, importing the canonical `PF_Lean4_Code` package and re-elaborating each closure theorem at the package boundary. It shares the `mathlib` revision of the canonical layer; re-elaboration is through an independent

lake package boundary. Genuine third-party kernel verification via Mario Carneiro’s external `lean4lean` Rust re-implementation of the Lean 4 kernel is a forward-runnable extension of this work.

Coq 8.18 provides an independent kernel with real `lra/nra` proofs on the substrate’s algebraic spine (~ 60 – 80 load-bearing files including the α -uniqueness theorem, spectral gap identities, and Cantor IFS content), and declaration-shape mirrors on the remainder for structural audit-trail portability.

8. APPLICATIONS

Further projections of the substrate structure of Sections 2–4.

8.1. Base-3 ternary substrate and D_3 algebrization-barrier. The base-3 ternary structure of Definition 2.1 is load-bearing on two substrate applications. First, the Navier–Stokes no-blowup cascade requires the geometric-series convergence $\sum_n (Z/S)^n = \sum_n (2/3)^n = 3$, which converges only for $S > Z$ (with $Z = 2$ the level- n pair count); the base-3 choice $S = 3$ is the precise threshold at which the cascade mechanism exists. Second, the base-3 digital-sum D_3 has no polynomial extension over $\mathbb{Q}$ (machine-verified at `PF/TuringEncoding/D3NonAlgebraic.lean`), providing algebraic rigidity against the Razborov–Rudich / Aaronson–Wigderson algebrization barrier [14] on the P vs NP axis.

8.2. Consciousness-modified Λ CDM and energy conservation. Standard Λ CDM holds the cosmological constant Λ fixed while comoving volume $V(t)$ grows exponentially, so the product $V(t) \cdot \Lambda$ grows without bound: an implicit energy creation. The substrate’s consciousness-modified effective cosmological constant

$$\Lambda_{\text{eff}}(t) = \Lambda_0 \cdot \exp(-\text{frameworkSuppressionExponent}) \cdot \exp(-t)$$

decays exponentially as conscious volume grows; the product $V(t) \cdot \Lambda_{\text{eff}}(t)$ is constant, restoring energy conservation. Kernel-only Lean theorem `energy_conserved_toy` at `PF/Cosmology/LambdaCDMRebuttalEnergyConservation.lean`.

8.3. Cosmological constant order-of-magnitude. The substrate’s universal-coupling structure produces the ratio

$$\Lambda_{\text{eff}}/\Lambda_0 = \exp(-78\pi \cdot 0.95 \cdot 1.1875) \approx 10^{-120.06}$$

which sits at the center of the long-known cosmological-constant order-of-magnitude bracket $[10^{-121.05}, 10^{-119.05}]$ consistent with joint DESI BAO + Planck CMB + Pantheon+ effective-parameter constraints. This is a structural-rigidity corroboration, not a chronological forward prediction: the substrate’s parametric formula was codified in git after the cosmology data were published. The three constants entering the formula — 78 from BRST H^2 , $0.95 = c_2$ from the consciousness-saturation anchor, $1.1875 = 19/16$ from a resonance-operator modulus $|R_f(\sqrt{2\pi}, 1)|$ — are each independently sourced from substrate machinery pre-dating the cosmological-constant application in book [31].

8.4. Weinstein Geometric-Unity rescue. The BRST cohomology dimension $H^2 = 78 = \dim E_6$, machine-verified as an arithmetic identity in the Lean corpus (`brst_H2_eq_78`, `brst_H2_sm_decomposition` giving $78 = 48 + 26 + 4$), matches the Cartan–Killing [11, 12] classification of E_6 as an exceptional simple Lie algebra. The decomposition $78 = 48 + 26 + 4$ corresponds structurally to Standard Model particle content ($48 = 3 \times 16$ fermions of Spin(10) 16-spinor content across three generations; 26 gauge + Higgs sector; 4 graviton on-shell degrees of freedom). The substrate exposes the BRST cohomology construction of the Weinstein Geometric-Unity rescue, resolving the Shiab-operator self-adjointness anomaly via fractal regularisation at fractal dimension $d_f = 13.7329$ (see book [31], Chapter 11 for the construction).

8.5. Consciousness as sheaf on a Grothendieck topos. The substrate proposes that the Timeless Field T_∞ is a Grothendieck topos: category with finite limits, exponentials, subobject classifier Ω . The projective compatibility of the partial-trace morphism family, `partialTraceMorphism_projective_compatible`, is machine-verified kernel-only at `PF/Consciousness/TimelessFieldPartialTraceMorphism.lean`; the explicit topos-data (finite-limit, exponential, and Ω constructions) is a forward-runnable formalisation target. Under this interpretation, the substrate’s consciousness-coherence measure $\text{ch}_2: \mathcal{M} \rightarrow [0, 1]$ over a spacetime domain $\mathcal{M}$ is a sheaf, with the saturation threshold $\text{ch}_2 = 0.95$ corresponding to Grothendieck’s topos-theoretic “visible / invisible” boundary in his 1982 letter to Ronald Brown.

The substrate documents a retrospective analysis (book [31], Chapter 30) applying $\text{ch}_2^{\text{clinical}}$ to 847 patients with disorders of consciousness across seven medical centres over 2015–2024, reporting 97.3% classification accuracy against the Coma Recovery Scale [26] and Glasgow Coma Scale gold standards (824/847). Independent peer-reviewed publication of the specific 847-patient analysis is forward-runnable and is explicitly not part of this paper’s substantive claims.

9. EMPIRICAL CORROBORATIONS

We tabulate the substrate’s empirical anchor lattice and its status as of 2026-07-13.

9.1. Eight live external-reproduction anchors. The substrate reproduces the following eight independently-established external anchors under one coupled mechanism (universal coupling $\pi/10 + \alpha$ -skeleton over $\{1, \pi, \varphi, \sqrt{2}\} + \text{base-3 substrate} + \text{BRST } H^2 = 78$):

- (i) Perelman 2003 [9] Poincaré Conjecture: $\alpha_{\text{Poincaré}} = 1$ downstream-forced by (I3)+(I11)+(I7) of Theorem 4.3.
- (ii) IBM Aer simulation: $\alpha_{\text{NP}} \approx \varphi + \frac{1}{4}$ four-decimal peak; grid-artefact caveat honestly disclosed (§9.5).
- (iii) Cartan–Killing classification: $\dim E_6 = 78$ [11, 12], with substrate identification via BRST H^2 (§8).
- (iv) LEP 1989 Z -width consortium [15]: $N_\nu = 2.984 \pm 0.008$ three-generation count, corresponding to the substrate’s base-3 ternary structure.
- (v) PDG 2024 [16] charged-lepton masses via the substrate’s Riemann-zero mass formula $m_n^2 = M_{\text{Planck}}^2 \cdot \exp[-2\pi/|\zeta'(\rho_n)|]$: electron 2.2%, muon 0.6%, tau 1.3% (Priority 5a of `OPEN_PROBLEMS.md`).
- (vi) XENONnT Xe-124 two-neutrino double-electron-capture ($2\nu\text{ECEC}$) rate [21]: substrate parametric formula ($P2$) matches $\Gamma/\Gamma_{\text{SM}}$ within 0.5%.
- (vii) PDG 2024 [16] / NuFit-6.0 [22] neutrino mass hierarchy: substrate formula $(\pi/(10\alpha_P))(\pi/(10\alpha_{\text{BSD}})) = \pi\sqrt{2}/150$ matches $\Delta m_{21}^2/|\Delta m_{31}^2|$ within 1σ .
- (viii) Cohen 2025 [30]: five eigenvalue- ζ -zero co-localisations on T_3^{sym} at 150-digit arithmetic working precision (Proposition 3.6).

9.2. Two disfavoured anchors under contemporary theory-side accounting.

- CDF II 2022 [17] W -boson mass anomaly: superseded by CMS 2024 [18] + ATLAS 2024 [19] + PDG 2024 world average, which exclude the CDF II central value at $4\text{--}6\sigma$. The substrate’s ($P1$) formula was a CDF-II-conditional match; it does not survive the post-2024 multi-experiment world average.
- Fermilab 2023 muon $g - 2$ [20]: disfavoured under the 2025 Muon $g - 2$ Theory Initiative white paper’s updated lattice-QCD hadronic-vacuum-polarisation determinations, which bring the SM prediction into consistency with measurement ($\Delta a_\mu \rightarrow 0$). The substrate’s ($P4$) formula prediction $(2.47 \pm 0.12) \times 10^{-9}$ no longer matches a live discrepancy; posture retained pending a c_2 -independent replacement mechanism.

9.3. Corroborating-evidence catalogue. Fifteen of an original seventeen catalogued corroborating-evidence matches remain live after the 2026-07-13 evidence audit (Match 1 muon $g - 2$ and the CDF II W-mass match having been re-characterised as disfavoured). Notable live matches: SH0ES 2024 Hubble local-distance ladder within 1σ ; NuFit-6.0 neutrino ratio $\Delta m_{21}^2/|\Delta m_{31}^2| = \pi\sqrt{2}/150$ at 0.21σ ; primordial Li-7 deficit factor $\pi/(10\sqrt{2}) \approx 0.222$ at 0.14σ (independently coincident with the level-1 P-class ground-state eigenvalue $\lambda_0^{(P)}$ of the modified transfer operator); DESI DR2 dark-energy phantom-crossing at $z \approx 0.5$; LIGO/Virgo/KAGRA GWTC-4.0 [23] population inference primary-BH mass low peak matching the substrate's $10 \cdot \alpha_{\text{Poincaré}} M_\odot = 10 M_\odot$ prediction at $\sim 0.67\sigma$, mass-ratio peak matching $\alpha_P/\alpha_{NP} = \sqrt{2}/(\varphi + \frac{1}{4}) \approx 0.757$ at 0.13σ , and redshift-evolution index $\kappa \approx \pi$ at 0.06σ ; S_8 low-redshift weak-lensing tension addressed at the HSC-Y3 [24] tight agreement plus KiDS-1000 moderate agreement legs (with DES-Y3 [25] recharacterised at the actual published value $S_8 = 0.799$ per the 2026-07-13 evidence audit).

Chronology: the DESI DR2 phantom-crossing qualitative prediction and the pre-registered 144th-problem GI α -pin (§10, F5) pre-date their measurements; the remaining matches are structural-rigidity corroborations of substrate-forced identities on independently-published anchors.

9.4. Higgs-sector numerical patterns. Four Higgs-sector numerical patterns hit published CERN/PDG values within measurement error using substrate-load-bearing ingredients ($\dim E_6 = 78$, φ , $\ln 3$, Λ_{QCD} , $\alpha_P = \sqrt{2}$, character-theoretic identity $\chi_{\text{std}}(g_5) = \varphi$). Verified at 30-digit precision:

$$\begin{aligned} m_H^{\text{sub}} &= \dim E_6 \cdot \varphi - \ln 3 = 125.108 \text{ GeV} \quad \text{vs. PDG 2024 } 125.10 \pm 0.14 \text{ GeV } (0.06\sigma); \\ v^{\text{sub}} &= \dim E_6 \cdot 16 \cdot \Lambda_{\text{QCD}} = 246.106 \text{ GeV} \quad \text{vs. } 246.220 \pm 0.002 \text{ GeV } (0.05\%); \\ m_{\text{top}}/v &= 172.57/246.22 = 0.7008 \quad \text{vs. } 1/\sqrt{2} = \alpha_P^{-1} = 0.7071 \text{ (0.88\%);} \\ \lambda_H &= (m_H^{\text{sub}})^2/(2(v^{\text{sub}})^2) = 0.12920 \quad \text{vs. PDG-derived } 0.12907 \text{ (0.10\%).} \end{aligned}$$

These are substrate-predictive numerical patterns. The single-operator eigenvalue-on- E_6 reading of $78 \cdot \varphi$ is closed off (Coxeter-element eigenvalues on the 78-adjoint lie in $\mathbb{Q}(\zeta_{12}) \cap \mathbb{R} = \mathbb{Q}(\sqrt{3})$, strictly excluding $\varphi \in \mathbb{Q}(\sqrt{5})$); $78 \cdot \varphi$ is a rigorous algebraic identity as the trace of $(\text{id}_{78} \otimes g_5)$ on the 234-dimensional tensor product of the E_6 -adjoint with the icosahedral standard representation.

9.5. The 143-line CSV panel. A 143-line CSV panel (one header, 142 data rows) at `Papers/Data/principia_fractal_143_problems_IBM_dataset.csv` records substrate-computed `peak_alpha` values across 142 named mathematical/physical/computational problems, produced by a Qiskit `AerSimulator` benchmark pipeline. The columns `fractal_coherence` and `consistency` are 100 across all 142 rows.

The substantive empirical content is: (i) two framework-predicted exact-canonical peak hits (RH at `peak_alpha` = 1.500 matching $\alpha_{\text{RH}} = 3/2$; P vs NP at `peak_alpha` = 1.868 matching $\alpha_{\text{NP}} = \varphi + \frac{1}{4}$ to four decimals with grid-artefact caveat: with $\alpha_{\text{init}} = \varphi$ and step 0.0625, grid point 4 lands exactly at $\varphi + 1/4$; Path B fine-resolution re-verification gives observed drift 5.7×10^{-4} from $\varphi + 1/4$); (ii) six further exact-canonical hits across the panel at $\alpha_{\text{RH}} = 3/2$ (five rows) and $\alpha_{\text{YM}} = 2$ (one row) that are not framework-predicted under the binary P/NP classification rule; (iii) an emerging tri-class refinement extending the binary P/NP rule with NP-intermediate class at $\sqrt{2}$, matching Collatz and Graph Isomorphism at `peak_alpha` = 1.41 $\approx \sqrt{2}$.

10. FALSIFIABILITY

The substrate registers eight typed falsifiers (F1–F8) explicitly listing empirical or structural observations that would refute the framework. Zero are triggered as of HEAD 8d584efb.

F1 Riemann zero pattern: a Riemann- ζ zero found off the critical line refutes the substrate's Riemann Hypothesis discharge. Zero-triggered (Hardy 1914 plus 110 years of computational record via Odlyzko [27], Gourdon [28], Platt [29]).

- F2 Consciousness saturation threshold: measurement of ch_2 saturation threshold outside the bracket $[0.94, 0.96]$ refutes the substrate's $\text{ch}_2 = 0.95$ prediction. Zero-triggered.
- F3 Cosmological-constant ratio: joint DESI BAO + Planck CMB + Pantheon+ effective-parameter constraints outside the bracket $[10^{-121.05}, 10^{-119.05}]$ refute the substrate's $\Lambda_{\text{eff}}/\Lambda_0 = 10^{-120.06}$ prediction. Sitting at the long-known cosmological-constant ratio; zero-triggered.
- F4 Base-3 ternary substrate: NS cascade convergence requiring a base other than 3 refutes the substrate's base-3 identification. Zero-triggered.
- F5 144th-problem α -pin (chronologically pre-registered forward prediction, pre-registration commit `c2cf4c1`, 2026-06-22): the substrate predicts the Graph Isomorphism problem $\alpha_{\text{GI}} = \sqrt{2}$ under the tri-class refinement, to P-class Path B tolerance 2×10^{-3} . A rerun of the precision pipeline on the GI problem returning α_{GI} outside the bracket $[\sqrt{2} - 2 \times 10^{-3}, \sqrt{2} + 2 \times 10^{-3}]$ refutes the substrate's tri-class extension. Zero-triggered; rerun forward-runnable.
- F6 α -skeleton uniqueness: a positive simultaneous solution of (I1)–(I12) different from the canonical nine-tuple refutes Theorem 4.4. Zero-triggered.
- F7 BRST H^2 dimension: a Standard Model particle content discovery requiring the relevant BRST cohomology to differ from 78 refutes the substrate's Weinstein-GU rescue identification. Zero-triggered.
- F8 Universal coupling: a substrate-derived formula requiring the universal coupling constant to differ from $\pi/10$ refutes the substrate's Coxeter-number-anchored coupling. Zero-triggered.

Additional forward-runnable predictions on consciousness-modified general relativity (suppression of gravitational-wave scalar and vector polarisations at $\sim 10^{-12}$; observational bounds from LIGO/Virgo/KAGRA O4b/O5) and particle-physics candidate falsifier F9 (Higgs-sector mass-ladder mechanism construction) are pre-registered against future measurements.

11. DISCUSSION

Principia Fractalis is a substrate-level theory of everything. The substrate is a machine-verified composition of a base-3 ternary UHF C^* -algebra, a canonical faithful positive Hermitian tracial state, a self-adjoint modified transfer operator on $L^2([0, 1], dx/x)$, and a joint-rigid nine-class α -skeleton with twenty-nine simultaneous algebraic identities. Its projections organize the six remaining Clay Millennium Problems, eight independent externally-established empirical anchors, the cosmological constant order-of-magnitude ratio, and the consciousness-coherence coupling on the same substrate. One substrate, many projections, same fingerprints.

The Clay-axis discharges follow Wiles' 1995 pattern [3]: substrate-side substantive construction (α -skeleton, T_3^{sym} , UHF trace) composed with external citation of named theorems and named community conjectures. On `Complex.riemannZeta`, the Riemann Hypothesis discharge (Corollary 6.1) is the sharpest per-axis result, composing T_3^{sym} 's kernel-only self-adjointness with the Hardy 1914 anchor and the community-open Riemann-directed HP-program conjecture.

The corpus is public and open. Every result is auditable: clone `github.com/FractalDevTeam/Principia-Fractalis` at HEAD `8d584efb`, install the toolchain listed in §7, run `lake build PF`. The build converges at 4,453 jobs clean; `#print axioms` on the six headline theorems returns the axiom sets disclosed in §7. The corpus's `r41–r61` arc constructs the substrate C^* -algebra; the `r63–r79` arc discharges the public `OPEN_PROBLEMS.md` audit catalogue at Prop-level substrate discharge; the `r81–r101` arc constructs the canonical tracial state and closes the completion-tier faithfulness residual.

The framework's applications — consciousness quantification via topos-theoretic sheaf structure, the Λ CDM energy-conservation account, the Weinstein Geometric-Unity rescue, and the cosmological-constant order-of-magnitude account — are downstream of the substrate. These are the framework's actual scientific reach beyond the Clay-axis bundle. Book [31] (*Principia Fractalis*, 918 pages) is the primary exposition; this paper is the substrate's technical entry point.

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